Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
CSC301	Data Structures and Algorithms	3	1	--	--	3	1	--	4
		Examination Scheme							
			ISE1	MSE	ISE2	ESE	Total		
		Theory	20	--	20	60	100		

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:	
Course Outcomes	**CO1** Identify data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2 Explain data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3 Apply data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4 Analyze data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5 Design data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Module No.	Unit No.	Topics	Ref.	Hrs.
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Data Structures and Algorithms: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Data Structures and Algorithms. Pearson, 1st, 2024.
Department Faculty Board. Data Structures and Algorithms Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3